

Pre-estimation (step 1)

ADF and PP test

for ln_total_tbr

Null Hypothesis: LN_TOTAL_TBR has a unit root

Exogenous: Constant

Lag Length: 1 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.448472	0.5497
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation

Dependent Variable: D(LN_TOTAL_TBR)

Method: Least Squares

Date: 06/07/26 Time: 09:16

Sample (adjusted): 1982 2024

Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_TOTAL_TBR(-1)	-0.071274	0.049206	-1.448472	0.1553
D(LN_TOTAL_TBR(-1))	0.319506	0.163768	1.950962	0.0581
C	-0.049683	0.034545	-1.438206	0.1582
R-squared	0.110603	Mean dependent var	-0.006849	
Adjusted R-squared	0.066133	S.D. dependent var	0.085293	
S.E. of regression	0.082425	Akaike info criterion	-2.086647	
Sum squared resid	0.271754	Schwarz criterion	-1.963773	
Log likelihood	47.86291	Hannan-Quinn criter.	-2.041335	
F-statistic	2.487136	Durbin-Watson stat	1.735919	
Prob(F-statistic)	0.095922			

Null Hypothesis: LN_TOTAL_TBR has a unit root

Exogenous: Constant

Bandwidth: 0 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*
Phillips-Perron test statistic	-1.151196	0.6869
Test critical values: 1% level	-3.588509	
5% level	-2.929734	
10% level	-2.603064	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.006846
HAC corrected variance (Bartlett kernel)	0.006846

Phillips-Perron Test Equation
 Dependent Variable: D(LN_TOTAL_TBR)
 Method: Least Squares
 Date: 06/07/26 Time: 09:16
 Sample (adjusted): 1981 2024
 Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_TOTAL_TBR(-1)	-0.056565	0.049136	-1.151196	0.2562
C	-0.045627	0.034686	-1.315429	0.1955
R-squared	0.030588	Mean dependent var	-0.008500	
Adjusted R-squared	0.007507	S.D. dependent var	0.085004	
S.E. of regression	0.084685	Akaike info criterion	-2.055377	
Sum squared resid	0.301202	Schwarz criterion	-1.974277	
Log likelihood	47.21829	Hannan-Quinn criter.	-2.025301	
F-statistic	1.325251	Durbin-Watson stat	1.349174	
Prob(F-statistic)	0.256163			

Null Hypothesis: D(LN_TOTAL_TBR) has a unit root
 Exogenous: Constant
 Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-4.476147	0.0008
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
 Dependent Variable: D(LN_TOTAL_TBR,2)
 Method: Least Squares
 Date: 06/07/26 Time: 09:16
 Sample (adjusted): 1982 2024
 Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_TOTAL_TBR(-1))	-0.727850	0.162606	-4.476147	0.0001

C	-0.003181	0.012924	-0.246107	0.8068
R-squared	0.328264	Mean dependent var	0.006629	
Adjusted R-squared	0.311880	S.D. dependent var	0.100685	
S.E. of regression	0.083521	Akaike info criterion	-2.082036	
Sum squared resid	0.286008	Schwarz criterion	-2.000120	
Log likelihood	46.76378	Hannan-Quinn criter.	-2.051828	
F-statistic	20.03589	Durbin-Watson stat	1.707648	
Prob(F-statistic)	0.000059			

Null Hypothesis: D(LN_TOTAL_TBR) has a unit root

Exogenous: Constant

Bandwidth: 3 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*
Phillips-Perron test statistic	-4.321666	0.0013
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.006651
HAC corrected variance (Bartlett kernel)	0.005778

Phillips-Perron Test Equation

Dependent Variable: D(LN_TOTAL_TBR,2)

Method: Least Squares

Date: 06/07/26 Time: 09:17

Sample (adjusted): 1982 2024

Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_TOTAL_TBR(-1))	-0.727850	0.162606	-4.476147	0.0001
C	-0.003181	0.012924	-0.246107	0.8068
R-squared	0.328264	Mean dependent var	0.006629	
Adjusted R-squared	0.311880	S.D. dependent var	0.100685	
S.E. of regression	0.083521	Akaike info criterion	-2.082036	
Sum squared resid	0.286008	Schwarz criterion	-2.000120	
Log likelihood	46.76378	Hannan-Quinn criter.	-2.051828	
F-statistic	20.03589	Durbin-Watson stat	1.707648	
Prob(F-statistic)	0.000059			

for ln_reer

Null Hypothesis: LN_REER has a unit root
Exogenous: Constant
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.620392	0.4637
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(LN_REER)
Method: Least Squares
Date: 06/07/26 Time: 09:18
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_REER(-1)	-0.071604	0.044190	-1.620392	0.1130
D(LN_REER(-1))	0.351601	0.141679	2.481675	0.0174
C	0.144727	0.089703	1.613402	0.1145
R-squared	0.160781	Mean dependent var	-0.000465	
Adjusted R-squared	0.118820	S.D. dependent var	0.021926	
S.E. of regression	0.020582	Akaike info criterion	-4.861605	
Sum squared resid	0.016944	Schwarz criterion	-4.738731	
Log likelihood	107.5245	Hannan-Quinn criter.	-4.816293	
F-statistic	3.831683	Durbin-Watson stat	2.092057	
Prob(F-statistic)	0.030027			

Null Hypothesis: D(LN_REER) has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-4.817806	0.0003
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation

Dependent Variable: D(LN_REER,2)
Method: Least Squares
Date: 06/07/26 Time: 09:18
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_REER(-1))	-0.686388	0.142469	-4.817806	0.0000
C	-0.000538	0.003200	-0.168115	0.8673
R-squared	0.361483	Mean dependent var	-0.000698	
Adjusted R-squared	0.345909	S.D. dependent var	0.025948	
S.E. of regression	0.020986	Akaike info criterion	-4.844540	
Sum squared resid	0.018057	Schwarz criterion	-4.762623	
Log likelihood	106.1576	Hannan-Quinn criter.	-4.814331	
F-statistic	23.21125	Durbin-Watson stat	2.027098	
Prob(F-statistic)	0.000020			

Null Hypothesis: LN_REER has a unit root
Exogenous: Constant
Bandwidth: 2 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*
Phillips-Perron test statistic	-1.141652	0.6908
Test critical values: 1% level	-3.588509	
5% level	-2.929734	
10% level	-2.603064	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.000486
HAC corrected variance (Bartlett kernel)	0.000733

Phillips-Perron Test Equation
Dependent Variable: D(LN_REER)
Method: Least Squares
Date: 06/07/26 Time: 09:18
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_REER(-1)	-0.041926	0.047400	-0.884509	0.3815
C	0.085574	0.096294	0.888678	0.3792
R-squared	0.018287	Mean dependent var	0.000455	
Adjusted R-squared	-0.005087	S.D. dependent var	0.022511	

S.E. of regression	0.022569	Akaike info criterion	-4.700122
Sum squared resid	0.021392	Schwarz criterion	-4.619023
Log likelihood	105.4027	Hannan-Quinn criter.	-4.670046
F-statistic	0.782357	Durbin-Watson stat	1.266278
Prob(F-statistic)	0.381457		

Null Hypothesis: D(LN_REER) has a unit root

Exogenous: Constant

Bandwidth: 2 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*
Phillips-Perron test statistic	-4.813292	0.0003
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.000420
HAC corrected variance (Bartlett kernel)	0.000417

Phillips-Perron Test Equation

Dependent Variable: D(LN_REER,2)

Method: Least Squares

Date: 06/07/26 Time: 09:18

Sample (adjusted): 1982 2024

Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_REER(-1))	-0.686388	0.142469	-4.817806	0.0000
C	-0.000538	0.003200	-0.168115	0.8673
R-squared	0.361483	Mean dependent var	-0.000698	
Adjusted R-squared	0.345909	S.D. dependent var	0.025948	
S.E. of regression	0.020986	Akaike info criterion	-4.844540	
Sum squared resid	0.018057	Schwarz criterion	-4.762623	
Log likelihood	106.1576	Hannan-Quinn criter.	-4.814331	
F-statistic	23.21125	Durbin-Watson stat	2.027098	
Prob(F-statistic)	0.000020			

for ln_nepal_gdp

Null Hypothesis: LN_NEPAL_GDP has a unit root

Exogenous: Constant

Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	0.386381	0.9801
Test critical values: 1% level	-3.588509	
5% level	-2.929734	
10% level	-2.603064	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
 Dependent Variable: D(LN_NEPAL_GDP)
 Method: Least Squares
 Date: 06/07/26 Time: 09:19
 Sample (adjusted): 1981 2024
 Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_NEPAL_GDP(-1)	0.004537	0.011741	0.386381	0.7012
C	-0.014471	0.116374	-0.124346	0.9016
R-squared	0.003542	Mean dependent var		0.030455
Adjusted R-squared	-0.020183	S.D. dependent var		0.031985
S.E. of regression	0.032306	Akaike info criterion		-3.982724
Sum squared resid	0.043835	Schwarz criterion		-3.901624
Log likelihood	89.61993	Hannan-Quinn criter.		-3.952648
F-statistic	0.149290	Durbin-Watson stat		1.785682
Prob(F-statistic)	0.701165			

Null Hypothesis: LN_NEPAL_GDP has a unit root
 Exogenous: Constant
 Bandwidth: 2 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*
Phillips-Perron test statistic	0.379838	0.9798
Test critical values: 1% level	-3.588509	
5% level	-2.929734	
10% level	-2.603064	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.000996
HAC corrected variance (Bartlett kernel)	0.001011

Phillips-Perron Test Equation
 Dependent Variable: D(LN_NEPAL_GDP)

Method: Least Squares
Date: 06/07/26 Time: 09:19
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_NEPAL_GDP(-1)	0.004537	0.011741	0.386381	0.7012
C	-0.014471	0.116374	-0.124346	0.9016
R-squared	0.003542	Mean dependent var	0.030455	
Adjusted R-squared	-0.020183	S.D. dependent var	0.031985	
S.E. of regression	0.032306	Akaike info criterion	-3.982724	
Sum squared resid	0.043835	Schwarz criterion	-3.901624	
Log likelihood	89.61993	Hannan-Quinn criter.	-3.952648	
F-statistic	0.149290	Durbin-Watson stat	1.785682	
Prob(F-statistic)	0.701165			

Null Hypothesis: D(LN_NEPAL_GDP) has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-5.917247	0.0000
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(LN_NEPAL_GDP,2)
Method: Least Squares
Date: 06/07/26 Time: 09:19
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_NEPAL_GDP(-1))	-0.903964	0.152768	-5.917247	0.0000
C	0.026587	0.006768	3.928237	0.0003
R-squared	0.460624	Mean dependent var	-0.001163	
Adjusted R-squared	0.447469	S.D. dependent var	0.043051	
S.E. of regression	0.032001	Akaike info criterion	-4.000719	
Sum squared resid	0.041986	Schwarz criterion	-3.918802	
Log likelihood	88.01545	Hannan-Quinn criter.	-3.970510	
F-statistic	35.01381	Durbin-Watson stat	1.940951	
Prob(F-statistic)	0.000001			

Null Hypothesis: D(LN_NEPAL_GDP) has a unit root
 Exogenous: Constant
 Bandwidth: 2 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*
Phillips-Perron test statistic	-5.904496	0.0000
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.000976
HAC corrected variance (Bartlett kernel)	0.000917

Phillips-Perron Test Equation
 Dependent Variable: D(LN_NEPAL_GDP,2)
 Method: Least Squares
 Date: 06/07/26 Time: 09:19
 Sample (adjusted): 1982 2024
 Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_NEPAL_GDP(-1))	-0.903964	0.152768	-5.917247	0.0000
C	0.026587	0.006768	3.928237	0.0003
R-squared	0.460624	Mean dependent var	-0.001163	
Adjusted R-squared	0.447469	S.D. dependent var	0.043051	
S.E. of regression	0.032001	Akaike info criterion	-4.000719	
Sum squared resid	0.041986	Schwarz criterion	-3.918802	
Log likelihood	88.01545	Hannan-Quinn criter.	-3.970510	
F-statistic	35.01381	Durbin-Watson stat	1.940951	
Prob(F-statistic)	0.000001			

for ln_india_tbr

Null Hypothesis: LN_INDIA_TBR has a unit root
 Exogenous: Constant
 Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.690101	0.4292
Test critical values: 1% level	-3.588509	
5% level	-2.929734	

10% level

-2.603064

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation

Dependent Variable: D(LN_INDIA_TBR)

Method: Least Squares

Date: 06/07/26 Time: 09:20

Sample (adjusted): 1981 2024

Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_INDIA_TBR(-1)	-0.104577	0.061876	-1.690101	0.0984
C	-0.079208	0.045513	-1.740318	0.0891
R-squared	0.063680	Mean dependent var	-0.007653	
Adjusted R-squared	0.041386	S.D. dependent var	0.113162	
S.E. of regression	0.110795	Akaike info criterion	-1.517874	
Sum squared resid	0.515576	Schwarz criterion	-1.436774	
Log likelihood	35.39323	Hannan-Quinn criter.	-1.487798	
F-statistic	2.856441	Durbin-Watson stat	1.717034	
Prob(F-statistic)	0.098419			

Null Hypothesis: D(LN_INDIA_TBR) has a unit root

Exogenous: Constant

Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-5.593083	0.0000
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation

Dependent Variable: D(LN_INDIA_TBR,2)

Method: Least Squares

Date: 06/07/26 Time: 09:20

Sample (adjusted): 1982 2024

Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_INDIA_TBR(-1))	-0.966172	0.172744	-5.593083	0.0000
C	-0.006883	0.017851	-0.385612	0.7018
R-squared	0.432782	Mean dependent var	0.007573	

Adjusted R-squared	0.418947	S.D. dependent var	0.151943
S.E. of regression	0.115821	Akaike info criterion	-1.428146
Sum squared resid	0.549995	Schwarz criterion	-1.346229
Log likelihood	32.70513	Hannan-Quinn criter.	-1.397938
F-statistic	31.28258	Durbin-Watson stat	1.810808
Prob(F-statistic)	0.000002		

Null Hypothesis: LN_INDIA_TBR has a unit root

Exogenous: Constant

Bandwidth: 4 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*
Phillips-Perron test statistic	-1.884175	0.3365
Test critical values: 1% level	-3.588509	
5% level	-2.929734	
10% level	-2.603064	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.011718
HAC corrected variance (Bartlett kernel)	0.016564

Phillips-Perron Test Equation

Dependent Variable: D(LN_INDIA_TBR)

Method: Least Squares

Date: 06/07/26 Time: 09:20

Sample (adjusted): 1981 2024

Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_INDIA_TBR(-1)	-0.104577	0.061876	-1.690101	0.0984
C	-0.079208	0.045513	-1.740318	0.0891
R-squared	0.063680	Mean dependent var	-0.007653	
Adjusted R-squared	0.041386	S.D. dependent var	0.113162	
S.E. of regression	0.110795	Akaike info criterion	-1.517874	
Sum squared resid	0.515576	Schwarz criterion	-1.436774	
Log likelihood	35.39323	Hannan-Quinn criter.	-1.487798	
F-statistic	2.856441	Durbin-Watson stat	1.717034	
Prob(F-statistic)	0.098419			

Null Hypothesis: D(LN_INDIA_TBR) has a unit root

Exogenous: Constant

Bandwidth: 3 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*
Phillips-Perron test statistic	-5.723091	0.0000
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.012791
HAC corrected variance (Bartlett kernel)	0.014713

Phillips-Perron Test Equation
 Dependent Variable: D(LN_INDIA_TBR,2)
 Method: Least Squares
 Date: 06/07/26 Time: 09:20
 Sample (adjusted): 1982 2024
 Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_INDIA_TBR(-1))	-0.966172	0.172744	-5.593083	0.0000
C	-0.006883	0.017851	-0.385612	0.7018
R-squared	0.432782	Mean dependent var		0.007573
Adjusted R-squared	0.418947	S.D. dependent var		0.151943
S.E. of regression	0.115821	Akaike info criterion		-1.428146
Sum squared resid	0.549995	Schwarz criterion		-1.346229
Log likelihood	32.70513	Hannan-Quinn criter.		-1.397938
F-statistic	31.28258	Durbin-Watson stat		1.810808
Prob(F-statistic)	0.000002			

for ln_row_tbr

Null Hypothesis: LN_ROW_TBR has a unit root
 Exogenous: Constant
 Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-1.662313	0.4430
Test critical values: 1% level	-3.588509	
5% level	-2.929734	
10% level	-2.603064	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation

Dependent Variable: D(LN_ROW_TBR)
Method: Least Squares
Date: 06/07/26 Time: 09:21
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_ROW_TBR(-1)	-0.135409	0.081458	-1.662313	0.1039
C	-0.093640	0.055011	-1.702212	0.0961
R-squared	0.061731	Mean dependent var	-0.007599	
Adjusted R-squared	0.039391	S.D. dependent var	0.126091	
S.E. of regression	0.123583	Akaike info criterion	-1.299421	
Sum squared resid	0.641454	Schwarz criterion	-1.218321	
Log likelihood	30.58726	Hannan-Quinn criter.	-1.269345	
F-statistic	2.763285	Durbin-Watson stat	1.708390	
Prob(F-statistic)	0.103897			

Null Hypothesis: D(LN_ROW_TBR) has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-6.121594	0.0000
Test critical values: 1% level	-3.592462	
5% level	-2.931404	
10% level	-2.603944	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(LN_ROW_TBR,2)
Method: Least Squares
Date: 06/07/26 Time: 09:21
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_ROW_TBR(-1))	-0.941600	0.153816	-6.121594	0.0000
C	-0.003621	0.019312	-0.187511	0.8522
R-squared	0.477533	Mean dependent var	0.001695	
Adjusted R-squared	0.464790	S.D. dependent var	0.172924	
S.E. of regression	0.126507	Akaike info criterion	-1.251635	
Sum squared resid	0.656170	Schwarz criterion	-1.169719	
Log likelihood	28.91016	Hannan-Quinn criter.	-1.221427	
F-statistic	37.47391	Durbin-Watson stat	1.926045	
Prob(F-statistic)	0.000000			

Null Hypothesis: LN_ROW_TBR has a unit root
 Exogenous: Constant
 Bandwidth: 5 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*
Phillips-Perron test statistic	-1.585191	0.4816
Test critical values: 1% level	-3.588509	
5% level	-2.929734	
10% level	-2.603064	

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.014579
HAC corrected variance (Bartlett kernel)	0.013384

Phillips-Perron Test Equation
 Dependent Variable: D(LN_ROW_TBR)
 Method: Least Squares
 Date: 06/07/26 Time: 09:21
 Sample (adjusted): 1981 2024
 Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_ROW_TBR(-1)	-0.135409	0.081458	-1.662313	0.1039
C	-0.093640	0.055011	-1.702212	0.0961
R-squared	0.061731	Mean dependent var		-0.007599
Adjusted R-squared	0.039391	S.D. dependent var		0.126091
S.E. of regression	0.123583	Akaike info criterion		-1.299421
Sum squared resid	0.641454	Schwarz criterion		-1.218321
Log likelihood	30.58726	Hannan-Quinn criter.		-1.269345
F-statistic	2.763285	Durbin-Watson stat		1.708390
Prob(F-statistic)	0.103897			

Null Hypothesis: D(LN_ROW_TBR) has a unit root
 Exogenous: Constant
 Bandwidth: 13 (Newey-West automatic) using Bartlett kernel

	Adj. t-Stat	Prob.*
Phillips-Perron test statistic	-6.503314	0.0000
Test critical values: 1% level	-3.592462	
5% level	-2.931404	

10% level

-2.603944

*MacKinnon (1996) one-sided p-values.

Residual variance (no correction)	0.015260
HAC corrected variance (Bartlett kernel)	0.006709

Phillips-Perron Test Equation
 Dependent Variable: D(LN_ROW_TBR,2)
 Method: Least Squares
 Date: 06/07/26 Time: 09:21
 Sample (adjusted): 1982 2024
 Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_ROW_TBR(-1))	-0.941600	0.153816	-6.121594	0.0000
C	-0.003621	0.019312	-0.187511	0.8522
R-squared	0.477533	Mean dependent var	0.001695	
Adjusted R-squared	0.464790	S.D. dependent var	0.172924	
S.E. of regression	0.126507	Akaike info criterion	-1.251635	
Sum squared resid	0.656170	Schwarz criterion	-1.169719	
Log likelihood	28.91016	Hannan-Quinn criter.	-1.221427	
F-statistic	37.47391	Durbin-Watson stat	1.926045	
Prob(F-statistic)	0.000000			

Stage 1: Linear ARDL baseline for aggregate form

Dependent Variable: LN_TOTAL_TBR
 Method: ARDL
 Date: 06/07/26 Time: 09:22
 Sample (adjusted): 1982 2024
 Included observations: 43 after adjustments
 Maximum dependent lags: 4 (Automatic selection)
 Model selection method: Schwarz criterion (SIC)
 Dynamic regressors (4 lags, automatic): LN_REER LN_NEPAL_GDP
 Fixed regressors: C
 Number of models evaluated: 100
 Selected Model: ARDL(2, 0, 0)
 Note: final equation sample is larger than selection sample

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
LN_TOTAL_TBR(-1)	1.050070	0.177288	5.922961	0.0000
LN_TOTAL_TBR(-2)	-0.385456	0.159297	-2.419729	0.0204

LN_REER	-0.381924	0.268566	-1.422088	0.1632
LN_NEPAL_GDP	-0.144595	0.060922	-2.373464	0.0228
C	1.989422	0.929139	2.141146	0.0387
R-squared	0.919128	Mean dependent var	-0.668253	
Adjusted R-squared	0.910615	S.D. dependent var	0.263826	
S.E. of regression	0.078877	Akaike info criterion	-2.132915	
Sum squared resid	0.236419	Schwarz criterion	-1.928124	
Log likelihood	50.85767	Hannan-Quinn criter.	-2.057394	
F-statistic	107.9699	Durbin-Watson stat	1.665641	
Prob(F-statistic)	0.000000			

*Note: p-values and any subsequent tests do not account for model selection.

ARDL Long Run Form and Bounds Test
Dependent Variable: D(LN_TOTAL_TBR)
Selected Model: ARDL(2, 0, 0)
Case 2: Restricted Constant and No Trend
Date: 06/07/26 Time: 09:22
Sample: 1980 2024
Included observations: 43

Conditional Error Correction Regression				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.989422	0.929139	2.141146	0.0387
LN_TOTAL_TBR(-1)*	-0.335386	0.124900	-2.685245	0.0107
LN_REER**	-0.381924	0.268566	-1.422088	0.1632
LN_NEPAL_GDP**	-0.144595	0.060922	-2.373464	0.0228
D(LN_TOTAL_TBR(-1))	0.385456	0.159297	2.419729	0.0204

* p-value incompatible with t-Bounds distribution.
** Variable interpreted as $Z = Z(-1) + D(Z)$.

Levels Equation Case 2: Restricted Constant and No Trend				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_REER	-1.138760	0.562763	-2.023516	0.0501
LN_NEPAL_GDP	-0.431131	0.092866	-4.642485	0.0000
C	5.931738	1.201722	4.936033	0.0000
EC = LN_TOTAL_TBR - (-1.1388*LN_REER -0.4311*LN_NEPAL_GDP + 5.9317)				

Null Hypothesis: No levels relationship				
F-Bounds Test				
Test Statistic	Value	Signif.	I(0)	I(1)

Asymptotic: n=1000				
F-statistic	2.009601	10%	2.63	3.35
k	2	5%	3.1	3.87
		2.5%	3.55	4.38
		1%	4.13	5
Finite Sample: n=45				
Actual Sample Size	43	10%	2.788	3.54
		5%	3.368	4.203
		1%	4.8	5.725
Finite Sample: n=40				
		10%	2.835	3.585
		5%	3.435	4.26
		1%	4.77	5.855

for india form

Dependent Variable: LN_INDIA_TBR

Method: ARDL

Date: 06/07/26 Time: 09:24

Sample (adjusted): 1981 2024

Included observations: 44 after adjustments

Maximum dependent lags: 4 (Automatic selection)

Model selection method: Schwarz criterion (SIC)

Dynamic regressors (4 lags, automatic): LN_REER LN_NEPAL_GDP

Fixed regressors: C

Number of models evaluated: 100

Selected Model: ARDL(1, 0, 0)

Note: final equation sample is larger than selection sample

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
LN_INDIA_TBR(-1)	0.849498	0.085731	9.908828	0.0000
LN_REER	-0.145494	0.245677	-0.592216	0.5570
LN_NEPAL_GDP	-0.034288	0.055926	-0.613093	0.5433
C	0.525409	0.651052	0.807016	0.4244
R-squared	0.836490	Mean dependent var	-0.691882	
Adjusted R-squared	0.824227	S.D. dependent var	0.267906	
S.E. of regression	0.112321	Akaike info criterion	-1.448410	
Sum squared resid	0.504637	Schwarz criterion	-1.286211	
Log likelihood	35.86502	Hannan-Quinn criter.	-1.388259	
F-statistic	68.21107	Durbin-Watson stat	1.686149	
Prob(F-statistic)	0.000000			

*Note: p-values and any subsequent tests do not account for model selection.

ARDL Long Run Form and Bounds Test
Dependent Variable: D(LN_INDIA_TBR)
Selected Model: ARDL(1, 0, 0)
Case 2: Restricted Constant and No Trend
Date: 06/07/26 Time: 09:24
Sample: 1980 2024
Included observations: 44

Conditional Error Correction Regression				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.525409	0.651052	0.807016	0.4244
LN_INDIA_TBR(-1)*	-0.150502	0.085731	-1.755505	0.0868
LN_REER**	-0.145494	0.245677	-0.592216	0.5570
LN_NEPAL_GDP**	-0.034288	0.055926	-0.613093	0.5433

* p-value incompatible with t-Bounds distribution.
** Variable interpreted as $Z = Z(-1) + D(Z)$.

Levels Equation Case 2: Restricted Constant and No Trend				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_REER	-0.966723	1.678519	-0.575938	0.5679
LN_NEPAL_GDP	-0.227824	0.303070	-0.751721	0.4566
C	3.491048	3.676083	0.949665	0.3480

EC = LN_INDIA_TBR - (-0.9667*LN_REER -0.2278*LN_NEPAL_GDP + 3.4910)

F-Bounds Test Null Hypothesis: No levels relationship

Test Statistic	Value	Signif.	I(0)	I(1)
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Asymptotic: n=1000				
F-statistic	0.962682	10%	2.63	3.35
k	2	5%	3.1	3.87
		2.5%	3.55	4.38
		1%	4.13	5

Finite Sample: n=45				
Actual Sample Size	44	10%	2.788	3.54
		5%	3.368	4.203
		1%	4.8	5.725

Finite

	Sample: n=40	
10%	2.835	3.585
5%	3.435	4.26
1%	4.77	5.855

for row form

Dependent Variable: LN_ROW_TBR
Method: ARDL
Date: 06/07/26 Time: 09:25
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments
Maximum dependent lags: 4 (Automatic selection)
Model selection method: Schwarz criterion (SIC)
Dynamic regressors (4 lags, automatic): LN_NEPAL_GDP LN_REER
Fixed regressors: C
Number of models evaluated: 100
Selected Model: ARDL(2, 0, 0)
Note: final equation sample is larger than selection sample

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
LN_ROW_TBR(-1)	0.592641	0.162579	3.645241	0.0008
LN_ROW_TBR(-2)	-0.431811	0.150222	-2.874489	0.0066
LN_NEPAL_GDP	-0.236083	0.058752	-4.018311	0.0003
LN_REER	-1.483590	0.454424	-3.264771	0.0023
C	4.821311	1.204260	4.003546	0.0003
R-squared	0.826051	Mean dependent var	-0.641064	
Adjusted R-squared	0.807741	S.D. dependent var	0.236786	
S.E. of regression	0.103824	Akaike info criterion	-1.583289	
Sum squared resid	0.409621	Schwarz criterion	-1.378498	
Log likelihood	39.04071	Hannan-Quinn criter.	-1.507768	
F-statistic	45.11369	Durbin-Watson stat	1.970100	
Prob(F-statistic)	0.000000			

*Note: p-values and any subsequent tests do not account for model selection.

ARDL Long Run Form and Bounds Test
Dependent Variable: D(LN_ROW_TBR)
Selected Model: ARDL(2, 0, 0)
Case 2: Restricted Constant and No Trend
Date: 06/07/26 Time: 09:25
Sample: 1980 2024
Included observations: 43

Conditional Error Correction Regression				
Variable	Coefficient	Std. Error	t-Statistic	Prob.

C	4.821311	1.204260	4.003546	0.0003
LN_ROW_TBR(-1)*	-0.839170	0.183509	-4.572909	0.0000
LN_NEPAL_GDP**	-0.236083	0.058752	-4.018311	0.0003
LN_REER**	-1.483590	0.454424	-3.264771	0.0023
D(LN_ROW_TBR(-1))	0.431811	0.150222	2.874489	0.0066

* p-value incompatible with t-Bounds distribution.

** Variable interpreted as $Z = Z(-1) + D(Z)$.

Levels Equation
Case 2: Restricted Constant and No Trend

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_NEPAL_GDP	-0.281328	0.049945	-5.632817	0.0000
LN_REER	-1.767925	0.291777	-6.059164	0.0000
C	5.745330	0.609782	9.421940	0.0000

$$EC = LN_ROW_TBR - (-0.2813 \cdot LN_NEPAL_GDP - 1.7679 \cdot LN_REER + 5.7453)$$

F-Bounds Test		Null Hypothesis: No levels relationship		
Test Statistic	Value	Signif.	I(0)	I(1)
Asymptotic: n=1000				
F-statistic	5.731072	10%	2.63	3.35
k	2	5%	3.1	3.87
		2.5%	3.55	4.38
		1%	4.13	5
Finite Sample: n=45				
Actual Sample Size	43	10%	2.788	3.54
		5%	3.368	4.203
		1%	4.8	5.725
Finite Sample: n=40				
		10%	2.835	3.585
		5%	3.435	4.26
		1%	4.77	5.855

Stage 2: ECM is also estimated please refer to stage 1 for it

Stage 3: NARDL for aggregate

Dependent Variable: LN_TOTAL_TBR
Method: ARDL
Date: 06/07/26 Time: 09:27
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments
Maximum dependent lags: 4 (Automatic selection)
Model selection method: Schwarz criterion (SIC)
Dynamic regressors (4 lags, automatic): LN_REER_POS LN_REER_NEG
LN_NEPAL_GDP
Fixed regressors: C
Number of models evaluated: 500
Selected Model: ARDL(2, 0, 0, 0)
Note: final equation sample is larger than selection sample

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
LN_TOTAL_TBR(-1)	1.040868	0.178610	5.827596	0.0000
LN_TOTAL_TBR(-2)	-0.368568	0.161599	-2.280756	0.0284
LN_REER_POS	0.218381	0.817911	0.266998	0.7910
LN_REER_NEG	-0.505104	0.313025	-1.613622	0.1151
LN_NEPAL_GDP	-0.324210	0.238988	-1.356596	0.1831
C	2.837877	2.183682	1.299584	0.2018
R-squared	0.920428	Mean dependent var	-0.668253	
Adjusted R-squared	0.909675	S.D. dependent var	0.263826	
S.E. of regression	0.079290	Akaike info criterion	-2.102610	
Sum squared resid	0.232618	Schwarz criterion	-1.856861	
Log likelihood	51.20612	Hannan-Quinn criter.	-2.011986	
F-statistic	85.59795	Durbin-Watson stat	1.651438	
Prob(F-statistic)	0.000000			

*Note: p-values and any subsequent tests do not account for model selection.

ARDL Long Run Form and Bounds Test
Dependent Variable: D(LN_TOTAL_TBR)
Selected Model: ARDL(2, 0, 0, 0)
Case 2: Restricted Constant and No Trend
Date: 06/07/26 Time: 09:27
Sample: 1980 2024
Included observations: 43

Conditional Error Correction Regression				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.837877	2.183682	1.299584	0.2018
LN_TOTAL_TBR(-1)*	-0.327699	0.125943	-2.601961	0.0133
LN_REER_POS**	0.218381	0.817911	0.266998	0.7910
LN_REER_NEG**	-0.505104	0.313025	-1.613622	0.1151

LN_NEPAL_GDP**	-0.324210	0.238988	-1.356596	0.1831
D(LN_TOTAL_TBR(-1))	0.368568	0.161599	2.280756	0.0284

* p-value incompatible with t-Bounds distribution.

** Variable interpreted as $Z = Z(-1) + D(Z)$.

Levels Equation
Case 2: Restricted Constant and No Trend

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_REER_POS	0.666405	2.588893	0.257409	0.7983
LN_REER_NEG	-1.541363	0.765323	-2.014005	0.0513
LN_NEPAL_GDP	-0.989351	0.772470	-1.280764	0.2082
C	8.660005	7.083414	1.222575	0.2292

$$EC = LN_TOTAL_TBR - (0.6664*LN_REER_POS - 1.5414*LN_REER_NEG - 0.9894 * LN_NEPAL_GDP + 8.6600)$$

F-Bounds Test Null Hypothesis: No levels relationship

Test Statistic	Value	Signif.	I(0)	I(1)
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Asymptotic:
n=1000

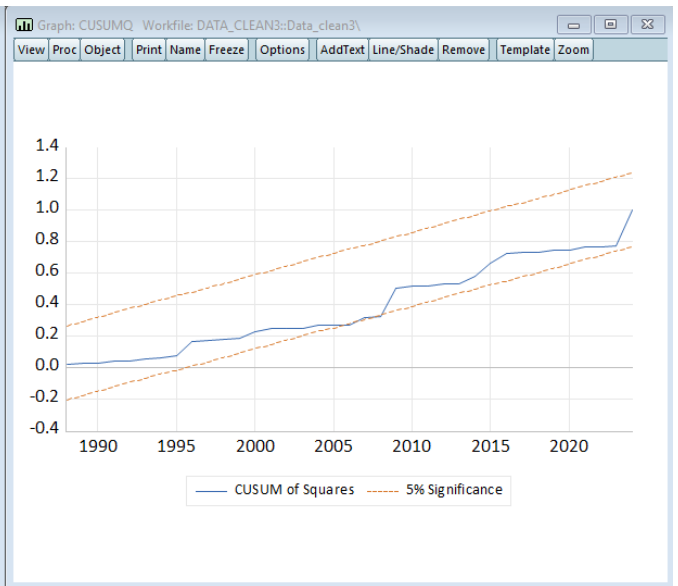
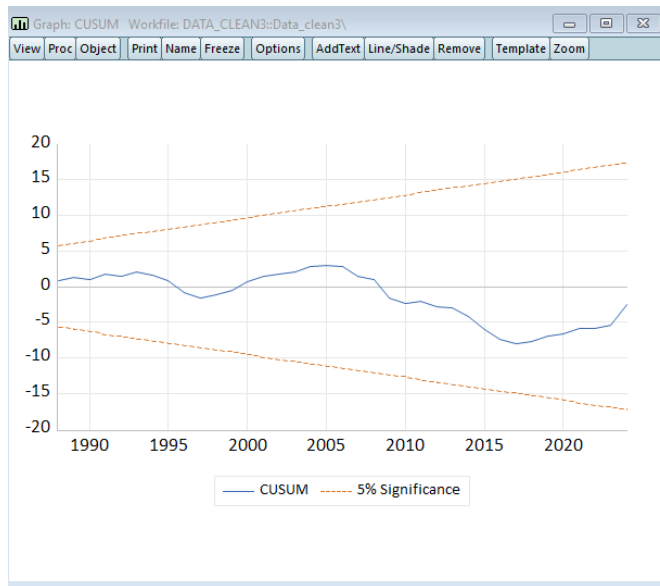
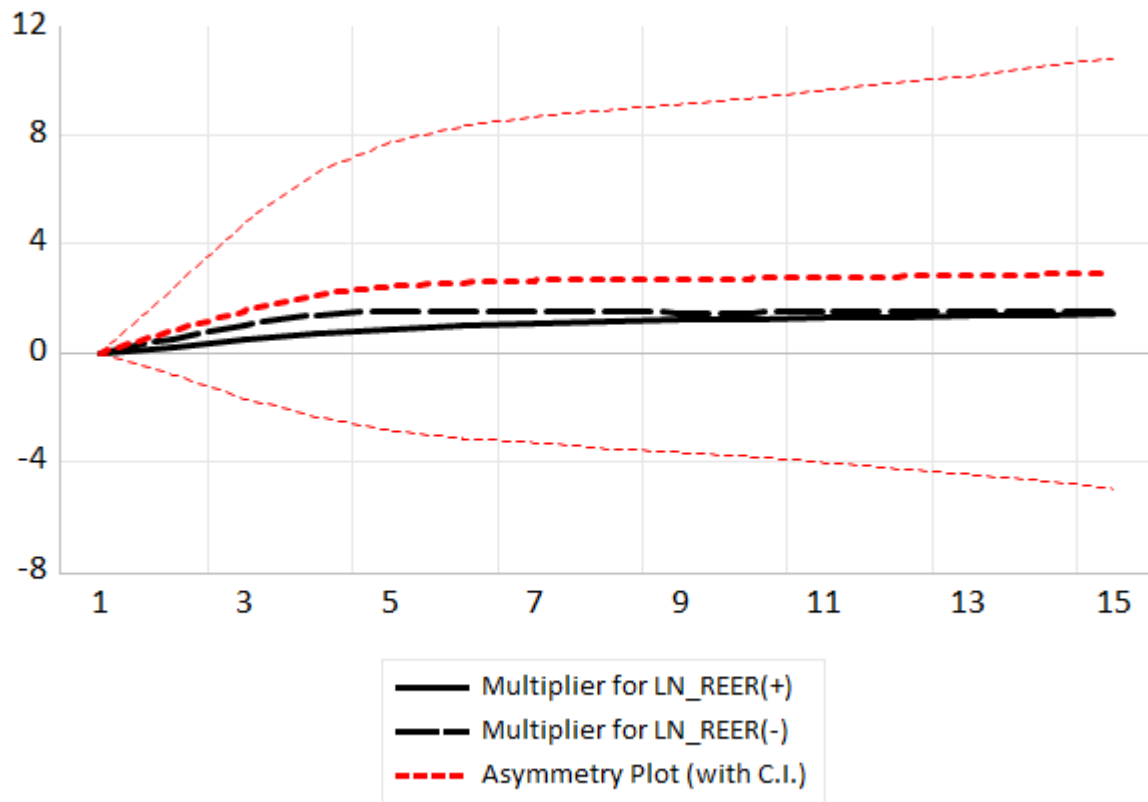
F-statistic	1.711860	10%	2.37	3.2
k	3	5%	2.79	3.67
		2.5%	3.15	4.08
		1%	3.65	4.66

Finite
Sample: n=45

Actual Sample Size	43	10%	2.56	3.428
		5%	3.078	4.022
		1%	4.27	5.412

Finite
Sample: n=40

10%	2.592	3.454
5%	3.1	4.088
1%	4.31	5.544



lr form

Dependent Variable: D(LN_TOTAL_TBR)

Method: Least Squares

Date: 06/07/26 Time: 09:30

Sample (adjusted): 1982 2024

Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.837877	2.183682	1.299584	0.2018
LN_TOTAL_TBR(-1)	-0.327699	0.125943	-2.601961	0.0133
LN_REER_POS	0.218381	0.817911	0.266998	0.7910
LN_REER_NEG	-0.505104	0.313025	-1.613622	0.1151
LN_NEPAL_GDP	-0.324210	0.238988	-1.356596	0.1831
D(LN_TOTAL_TBR(-1))	0.368568	0.161599	2.280756	0.0284
R-squared	0.238686	Mean dependent var	-0.006849	
Adjusted R-squared	0.135806	S.D. dependent var	0.085293	
S.E. of regression	0.079290	Akaike info criterion	-2.102610	
Sum squared resid	0.232618	Schwarz criterion	-1.856861	
Log likelihood	51.20612	Hannan-Quinn criter.	-2.011986	
F-statistic	2.320036	Durbin-Watson stat	1.651438	
Prob(F-statistic)	0.062631			

wald test

Wald Test:

Equation: LRFORM_AGGREGATE

Test Statistic	Value	df	Probability
t-statistic	0.777525	37	0.4418
F-statistic	0.604546	(1, 37)	0.4418
Chi-square	0.604546	1	0.4368

Null Hypothesis: C(3) = C(4)

Null Hypothesis Summary:

Normalized Restriction (= 0)	Value	Std. Err.
C(3) - C(4)	0.723484	0.930496

Restrictions are linear in coefficients.

Breusch-Godfrey test

Breusch-Godfrey Serial Correlation LM Test:

Null hypothesis: No serial correlation at up to 2 lags

F-statistic	0.008097	Prob. F(2,35)	0.9919
Obs*R-squared	0.019885	Prob. Chi-Square(2)	0.9901

Test Equation:

Dependent Variable: RESID

Method: ARDL

Date: 06/07/26 Time: 09:31

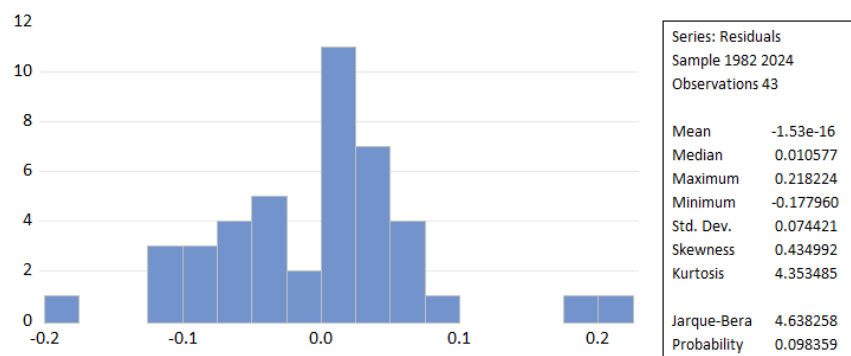
Sample: 1982 2024

Included observations: 43

Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_TOTAL_TBR(-1)	0.029737	0.768192	0.038710	0.9693
LN_TOTAL_TBR(-2)	-0.012531	0.522570	-0.023980	0.9810
LN_REER_POS	0.002148	0.952764	0.002255	0.9982
LN_REER_NEG	0.025103	0.580448	0.043247	0.9658
LN_NEPAL_GDP	0.012244	0.407606	0.030040	0.9762
C	-0.104655	3.640613	-0.028747	0.9772
RESID(-1)	-0.024248	0.793450	-0.030560	0.9758
RESID(-2)	-0.035840	0.340378	-0.105294	0.9167
R-squared	0.000462	Mean dependent var	-1.53E-16	
Adjusted R-squared	-0.199445	S.D. dependent var	0.074421	
S.E. of regression	0.081506	Akaike info criterion	-2.010050	
Sum squared resid	0.232511	Schwarz criterion	-1.682384	
Log likelihood	51.21607	Hannan-Quinn criter.	-1.889217	
F-statistic	0.002313	Durbin-Watson stat	1.660721	
Prob(F-statistic)	1.000000			

Jarque Bera



Ramsey reset

Ramsey RESET Test

Equation: NARDL

Omitted Variables: Squares of fitted values

Specification: LN_TOTAL_TBR LN_TOTAL_TBR(-1) LN_TOTAL_TBR(-2)

LN_REER_POS LN_REER_NEG LN_NEPAL_GDP C

	Value	df	Probability
t-statistic	0.657330	36	0.5151
F-statistic	0.432083	(1, 36)	0.5151
Likelihood ratio	0.513027	1	0.4738

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	0.002759	1	0.002759
Restricted SSR	0.232618	37	0.006287
Unrestricted SSR	0.229859	36	0.006385

LR test summary:

	Value
Restricted LogL	51.20612
Unrestricted LogL	51.46263

Unrestricted Test Equation:

Dependent Variable: LN_TOTAL_TBR

Method: Least Squares

Date: 06/07/26 Time: 09:34

Sample: 1982 2024

Included observations: 43

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_TOTAL_TBR(-1)	0.659447	0.607536	1.085445	0.2849
LN_TOTAL_TBR(-2)	-0.276374	0.214924	-1.285914	0.2067
LN_REER_POS	0.660077	1.063453	0.620693	0.5387
LN_REER_NEG	-0.427253	0.336955	-1.267983	0.2129
LN_NEPAL_GDP	-0.349213	0.243828	-1.432208	0.1607
C	2.979690	2.211186	1.347553	0.1862
FITTED^2	-0.294261	0.447661	-0.657330	0.5151
R-squared	0.921372	Mean dependent var	-0.668253	
Adjusted R-squared	0.908267	S.D. dependent var	0.263826	
		Akaike info		
S.E. of regression	0.079906	criterion	-2.068029	
Sum squared resid	0.229859	Schwarz criterion	-1.781322	
Log likelihood	51.46263	Hannan-Quinn criter.	-1.962301	
F-statistic	70.30876	Durbin-Watson stat	1.676728	
Prob(F-statistic)	0.000000			

Heteroscedasticity

Heteroskedasticity Test: Breusch-Pagan-Godfrey

Null hypothesis: Homoskedasticity

F-statistic	3.002165	Prob. F(5,37)	0.0226
Obs*R-squared	12.41021	Prob. Chi-Square(5)	0.0296
Scaled explained SS	15.40679	Prob. Chi-Square(5)	0.0088

Test Equation:

Dependent Variable: RESID^2

Method: Least Squares

Date: 06/07/26 Time: 09:34

Sample: 1982 2024

Included observations: 43

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.146519	0.248073	0.590630	0.5584
LN_TOTAL_TBR(-1)	-0.037280	0.020291	-1.837277	0.0742
LN_TOTAL_TBR(-2)	0.041387	0.018358	2.254427	0.0302
LN_REER_POS	0.100644	0.092917	1.083154	0.2857
LN_REER_NEG	0.059917	0.035561	1.684930	0.1004
LN_NEPAL_GDP	-0.014375	0.027150	-0.529457	0.5997
R-squared	0.288610	Mean dependent var	0.005410	
Adjusted R-squared	0.192476	S.D. dependent var	0.010024	
S.E. of regression	0.009008	Akaike info criterion	-6.452701	
Sum squared resid	0.003002	Schwarz criterion	-6.206952	
Log likelihood	144.7331	Hannan-Quinn criter.	-6.362076	
F-statistic	3.002165	Durbin-Watson stat	1.269618	
Prob(F-statistic)	0.022565			

for india

Dependent Variable: LN_INDIA_TBR

Method: ARDL

Date: 06/07/26 Time: 09:36

Sample (adjusted): 1981 2024

Included observations: 44 after adjustments

Maximum dependent lags: 4 (Automatic selection)

Model selection method: Schwarz criterion (SIC)

Dynamic regressors (4 lags, automatic): LN_REER_POS LN_REER_NEG

LN_NEPAL_GDP

Fixed regressors: C

Number of models evaluated: 500

Selected Model: ARDL(1, 0, 0, 0)

Note: final equation sample is larger than selection sample

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
LN_INDIA_TBR(-1)	0.860642	0.086379	9.963521	0.0000
LN_REER_POS	0.940574	1.091973	0.861353	0.3943
LN_REER_NEG	-0.370160	0.329756	-1.122527	0.2685
LN_NEPAL_GDP	-0.363182	0.327025	-1.110561	0.2736
C	3.238611	3.001717	1.078919	0.2873
R-squared	0.840744	Mean dependent var	-0.691882	
Adjusted R-squared	0.824411	S.D. dependent var	0.267906	
S.E. of regression	0.112262	Akaike info criterion	-1.429320	
Sum squared resid	0.491506	Schwarz criterion	-1.226571	
Log likelihood	36.44505	Hannan-Quinn criter.	-1.354131	
F-statistic	51.47236	Durbin-Watson stat	1.770795	
Prob(F-statistic)	0.000000			

*Note: p-values and any subsequent tests do not account for model selection.

ARDL Long Run Form and Bounds Test

Dependent Variable: D(LN_INDIA_TBR)

Selected Model: ARDL(1, 0, 0, 0)

Case 2: Restricted Constant and No Trend

Date: 06/07/26 Time: 09:36

Sample: 1980 2024

Included observations: 44

Conditional Error Correction Regression				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	3.238611	3.001717	1.078919	0.2873
LN_INDIA_TBR(-1)*	-0.139358	0.086379	-1.613320	0.1147
LN_REER_POS**	0.940574	1.091973	0.861353	0.3943
LN_REER_NEG**	-0.370160	0.329756	-1.122527	0.2685
LN_NEPAL_GDP**	-0.363182	0.327025	-1.110561	0.2736

* p-value incompatible with t-Bounds distribution.
 ** Variable interpreted as $Z = Z(-1) + D(Z)$.

Levels Equation
 Case 2: Restricted Constant and No Trend

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_REER_POS	6.749359	9.391471	0.718669	0.4766
LN_REER_NEG	-2.656187	2.912975	-0.911847	0.3675
LN_NEPAL_GDP	-2.606113	2.867012	-0.909000	0.3689
C	23.23958	26.20126	0.886964	0.3805

$$EC = LN_INDIA_TBR - (6.7494 \cdot LN_REER_POS - 2.6562 \cdot LN_REER_NEG - 2.6061 \cdot LN_NEPAL_GDP + 23.2396)$$

F-Bounds Test Null Hypothesis: No levels relationship

Test Statistic	Value	Signif.	I(0)	I(1)
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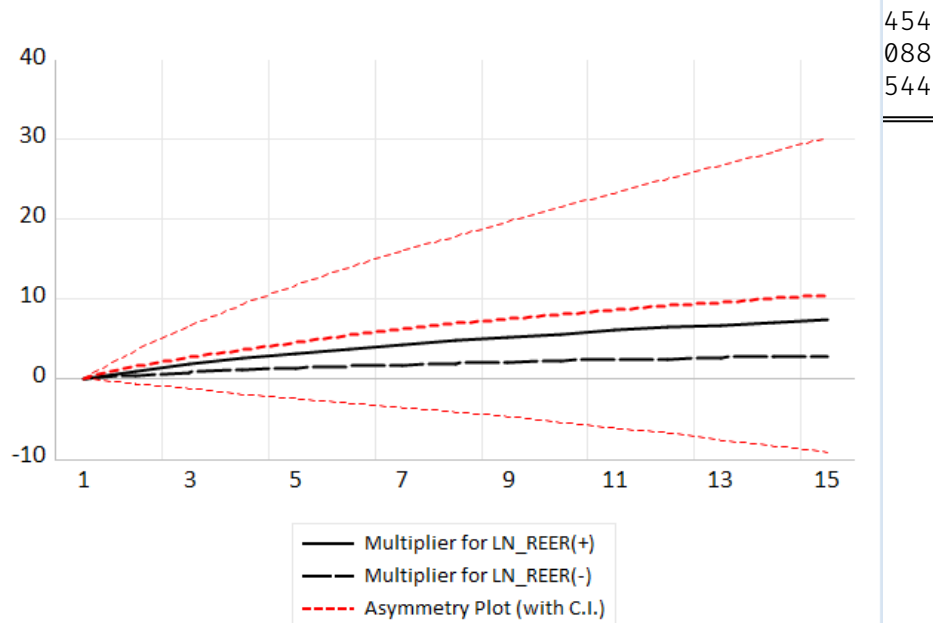
Asymptotic:
 n=1000

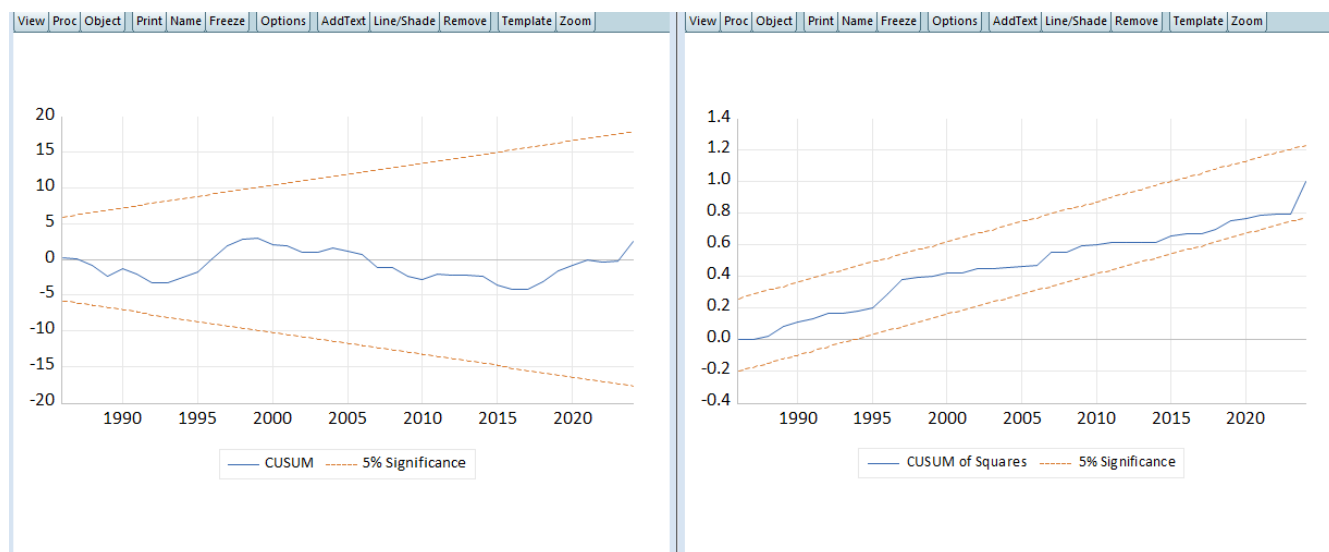
F-statistic	0.979332	10%	2.37	3.2
k	3	5%	2.79	3.67
		2.5%	3.15	4.08
		1%	3.65	4.66

Finite
 Sample: n=45

Actual Sample Size	44	10%	2.56	3.428
		5%	3.078	4.022
		1%	4.27	5.412

Finite
 Sample: n=40





Ir form

Dependent Variable: D(LN_INDIA_TBR)

Method: Least Squares

Date: 06/07/26 Time: 09:38

Sample (adjusted): 1981 2024

Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	3.238611	3.001717	1.078919	0.2873
LN_INDIA_TBR(-1)	-0.139358	0.086379	-1.613320	0.1147
LN_REER_POS	0.940574	1.091973	0.861353	0.3943
LN_REER_NEG	-0.370160	0.329756	-1.122527	0.2685
LN_NEPAL_GDP	-0.363182	0.327025	-1.110561	0.2736
R-squared	0.107392	Mean dependent var	-0.007653	
Adjusted R-squared	0.015842	S.D. dependent var	0.113162	
S.E. of regression	0.112262	Akaike info criterion	-1.429320	
Sum squared resid	0.491506	Schwarz criterion	-1.226571	

Log likelihood	36.44505	Hannan-Quinn criter.	-1.354131
F-statistic	1.173046	Durbin-Watson stat	1.770795
Prob(F-statistic)	0.337619		

wald test

Wald Test:
Equation: LRFORM_INDIA

Test Statistic	Value	df	Probability
t-statistic	1.020734	39	0.3137
F-statistic	1.041898	(1, 39)	0.3137
Chi-square	1.041898	1	0.3074

Null Hypothesis: C(3) = C(4)
Null Hypothesis Summary:

Normalized Restriction (= 0)	Value	Std. Err.
C(3) - C(4)	1.310734	1.284109

Restrictions are linear in coefficients.

Breusch-Godfrey test

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 2 lags

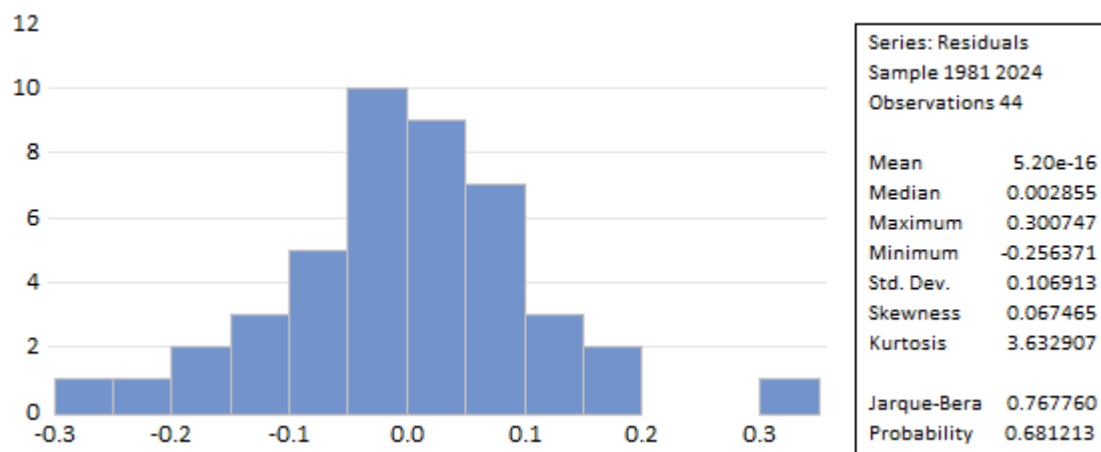
F-statistic	0.124901	Prob. F(2,37)	0.8830
Obs*R-squared	0.295069	Prob. Chi-Square(2)	0.8628

Test Equation:
Dependent Variable: RESID
Method: ARDL
Date: 06/07/26 Time: 09:39
Sample: 1981 2024
Included observations: 44
Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_INDIA_TBR(-1)	-0.032714	0.113950	-0.287094	0.7756
LN_REER_POS	-0.095149	1.133735	-0.083925	0.9336
LN_REER_NEG	0.014387	0.338654	0.042481	0.9663
LN_NEPAL_GDP	0.018675	0.337159	0.055390	0.9561
C	-0.185593	3.097406	-0.059919	0.9525
RESID(-1)	0.055995	0.215179	0.260224	0.7961
RESID(-2)	0.099601	0.208976	0.476613	0.6364

R-squared	0.006706	Mean dependent var	5.20E-16
Adjusted R-squared	-0.154369	S.D. dependent var	0.106913
S.E. of regression	0.114869	Akaike info criterion	-1.345140
Sum squared resid	0.488210	Schwarz criterion	-1.061292
Log likelihood	36.59308	Hannan-Quinn criter.	-1.239875
F-statistic	0.041634	Durbin-Watson stat	1.832458
Prob(F-statistic)	0.999660		

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Ramsay reset

Ramsey RESET Test
Equation: NARDL_INDIA
Omitted Variables: Squares of fitted values
Specification: LN_INDIA_TBR LN_INDIA_TBR(-1) LN_REER_POS
LN_REER_NEG LN_NEPAL_GDP C

	Value	df	Probability
t-statistic	0.549212	38	0.5861
F-statistic	0.301634	(1, 38)	0.5861
Likelihood ratio	0.347881	1	0.5553

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	0.003871	1	0.003871
Restricted SSR	0.491506	39	0.012603
Unrestricted SSR	0.487636	38	0.012833

LR test summary:

	Value
Restricted LogL	36.44505
Unrestricted LogL	36.61899

Unrestricted Test Equation:
 Dependent Variable: LN_INDIA_TBR
 Method: Least Squares
 Date: 06/07/26 Time: 09:40
 Sample: 1981 2024
 Included observations: 44

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_INDIA_TBR(-1)	1.144100	0.523425	2.185795	0.0351
LN_REER_POS	0.769263	1.145182	0.671739	0.5058
LN_REER_NEG	-0.457877	0.369094	-1.240543	0.2224
LN_NEPAL_GDP	-0.365054	0.330011	-1.106187	0.2756
C	3.329746	3.033501	1.097658	0.2793
FITTED^2	0.245068	0.446218	0.549212	0.5861
R-squared	0.841999	Mean dependent var	-0.691882	
Adjusted R-squared	0.821209	S.D. dependent var	0.267906	
		Akaike info		
S.E. of regression	0.113281	criterion	-1.391772	
Sum squared resid	0.487636	Schwarz criterion	-1.148474	
Log likelihood	36.61899	Hannan-Quinn criter.	-1.301545	
F-statistic	40.50085	Durbin-Watson stat	1.758399	
Prob(F-statistic)	0.000000			

Heteroscedasticity

Heteroskedasticity Test: Breusch-Pagan-Godfrey
 Null hypothesis: Homoskedasticity

F-statistic	0.499339	Prob. F(4,39)	0.7363
Obs*R-squared	2.143641	Prob. Chi-Square(4)	0.7094
Scaled explained SS	2.217081	Prob. Chi-Square(4)	0.6959

Test Equation:
 Dependent Variable: RESID^2
 Method: Least Squares
 Date: 06/07/26 Time: 09:40
 Sample: 1981 2024
 Included observations: 44

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.294942	0.502088	0.587431	0.5603
LN_INDIA_TBR(-1)	-0.017080	0.014448	-1.182106	0.2443
LN_REER_POS	0.092938	0.182651	0.508827	0.6137
LN_REER_NEG	-0.009432	0.055157	-0.171008	0.8651
LN_NEPAL_GDP	-0.031719	0.054701	-0.579869	0.5653
R-squared	0.048719	Mean dependent var	0.011171	

Adjusted R-squared	-0.048848	S.D. dependent var	0.018335
S.E. of regression	0.018778	Akaike info criterion	-5.005648
Sum squared resid	0.013751	Schwarz criterion	-4.802899
Log likelihood	115.1242	Hannan-Quinn criter.	-4.930459
F-statistic	0.499339	Durbin-Watson stat	1.323038
Prob(F-statistic)	0.736310		

for row

Dependent Variable: LN_ROW_TBR
Method: ARDL
Date: 06/07/26 Time: 09:44
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments
Maximum dependent lags: 4 (Automatic selection)
Model selection method: Akaike info criterion (AIC)
Dynamic regressors (4 lags, automatic): LN_NEPAL_GDP LN_REER_POS
LN_REER_NEG
Fixed regressors: C
Number of models evaluated: 500
Selected Model: ARDL(3, 0, 0, 0)
Note: final equation sample is larger than selection sample

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
LN_ROW_TBR(-1)	0.445805	0.142899	3.119719	0.0036
LN_ROW_TBR(-2)	-0.231161	0.145496	-1.588782	0.1211
LN_ROW_TBR(-3)	-0.141158	0.121469	-1.162085	0.2531
LN_NEPAL_GDP	-0.785940	0.275795	-2.849728	0.0073
LN_REER_POS	0.227442	0.780134	0.291542	0.7724
LN_REER_NEG	-1.667838	0.565767	-2.947926	0.0057
C	6.774036	2.467859	2.744904	0.0095
R-squared	0.898259	Mean dependent var	-0.629638	
Adjusted R-squared	0.880817	S.D. dependent var	0.227340	
S.E. of regression	0.078484	Akaike info criterion	-2.100822	
Sum squared resid	0.215593	Schwarz criterion	-1.811211	
Log likelihood	51.11726	Hannan-Quinn criter.	-1.994668	
F-statistic	51.50155	Durbin-Watson stat	1.822192	
Prob(F-statistic)	0.000000			

*Note: p-values and any subsequent tests do not account for model selection.

ARDL Long Run Form and Bounds Test
Dependent Variable: D(LN_ROW_TBR)
Selected Model: ARDL(3, 0, 0, 0)
Case 2: Restricted Constant and No Trend
Date: 06/07/26 Time: 09:45
Sample: 1980 2024
Included observations: 42

Conditional Error Correction Regression				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	6.774036	2.467859	2.744904	0.0095
LN_ROW_TBR(-1)*	-0.926513	0.193568	-4.786511	0.0000
LN_NEPAL_GDP**	-0.785940	0.275795	-2.849728	0.0073

LN_REER_POS**	0.227442	0.780134	0.291542	0.7724
LN_REER_NEG**	-1.667838	0.565767	-2.947926	0.0057
D(LN_ROW_TBR(-1))	0.372318	0.133112	2.797036	0.0083
D(LN_ROW_TBR(-2))	0.141158	0.121469	1.162085	0.2531

* p-value incompatible with t-Bounds distribution.

** Variable interpreted as $Z = Z(-1) + D(Z)$.

Levels Equation
Case 2: Restricted Constant and No Trend

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_NEPAL_GDP	-0.848278	0.257073	-3.299757	0.0022
LN_REER_POS	0.245482	0.849918	0.288830	0.7744
LN_REER_NEG	-1.800124	0.331132	-5.436280	0.0000
C	7.311323	2.358477	3.100020	0.0038

EC = LN_ROW_TBR - (-0.8483*LN_NEPAL_GDP + 0.2455*LN_REER_POS -1.8001
*LN_REER_NEG + 7.3113)

F-Bounds Test Null Hypothesis: No levels
relationship

Test Statistic	Value	Signif.	I(0)	I(1)
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Asymptotic:
n=1000

F-statistic	8.318168	10%	2.37	3.2
k	3	5%	2.79	3.67
		2.5%	3.15	4.08
		1%	3.65	4.66

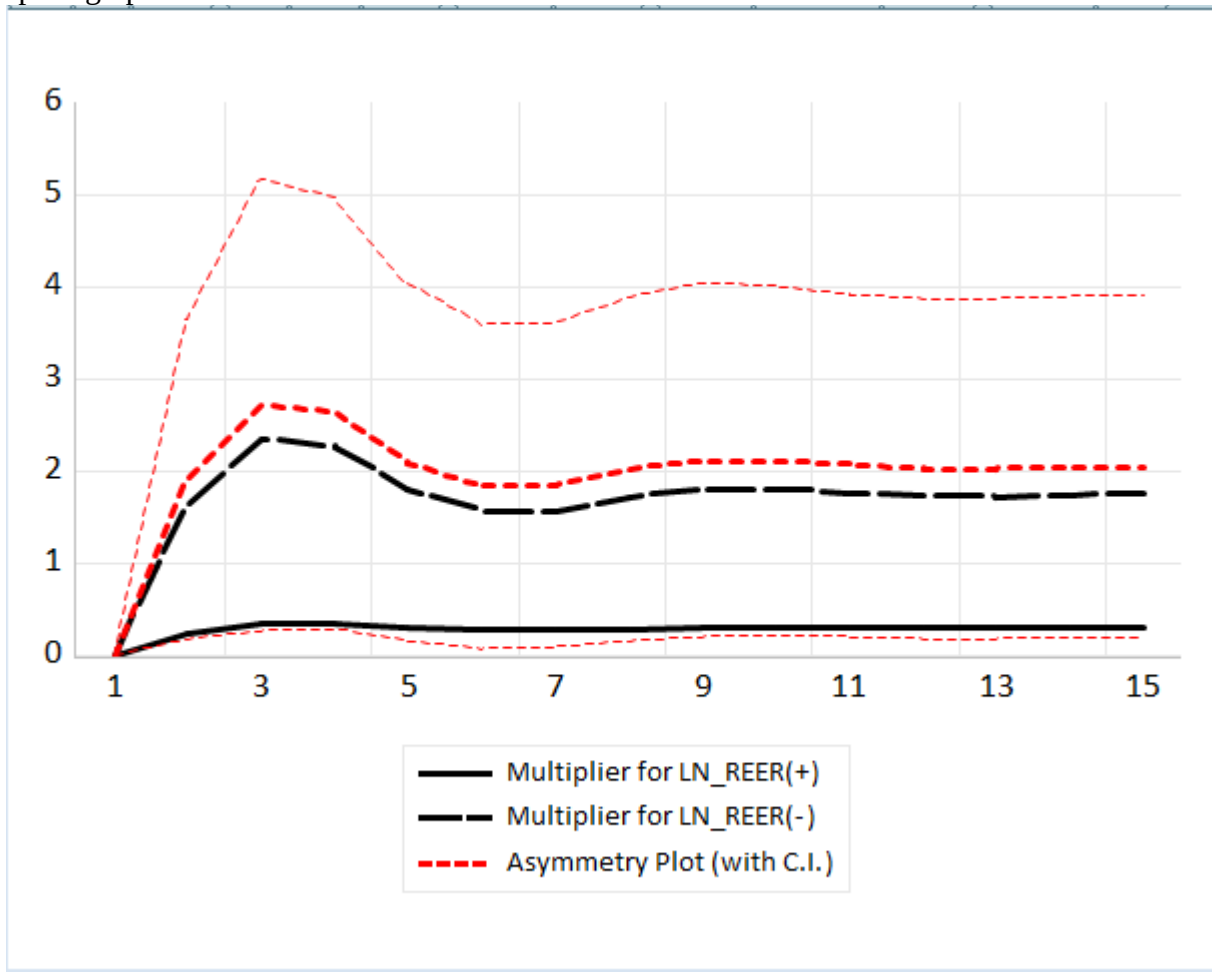
Finite
Sample: n=45

Actual Sample Size	42	10%	2.56	3.428
		5%	3.078	4.022
		1%	4.27	5.412

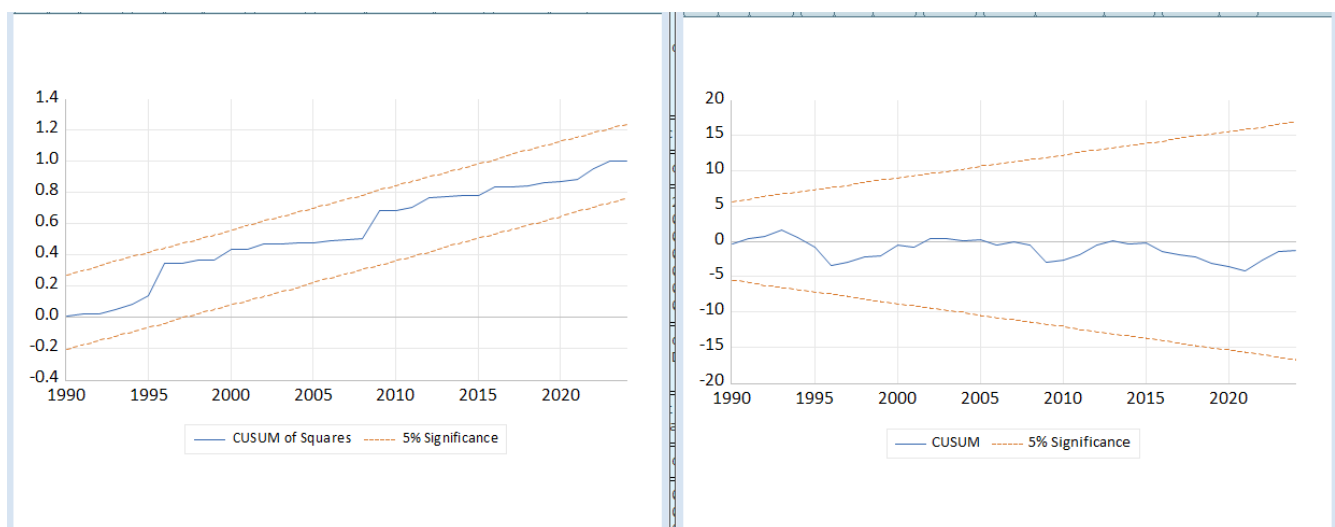
Finite
Sample: n=40

10%	2.592	3.454
5%	3.1	4.088
1%	4.31	5.544

Multiplier graph



CUMSUM and CUMSUMSQ



lr form

Dependent Variable: D(LN_ROW_TBR)
Method: Least Squares
Date: 06/07/26 Time: 09:46
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	6.774036	2.467859	2.744904	0.0095
LN_ROW_TBR(-1)	-0.926513	0.193568	-4.786511	0.0000
LN_NEPAL_GDP	-0.785940	0.275795	-2.849728	0.0073
LN_REER_POS	0.227442	0.780134	0.291542	0.7724
LN_REER_NEG	-1.667838	0.565767	-2.947926	0.0057
D(LN_ROW_TBR(-1))	0.372318	0.133112	2.797036	0.0083
D(LN_ROW_TBR(-2))	0.141158	0.121469	1.162085	0.2531
R-squared	0.570956	Mean dependent var	0.005343	
Adjusted R-squared	0.497405	S.D. dependent var	0.110707	
S.E. of regression	0.078484	Akaike info criterion	-2.100822	
Sum squared resid	0.215593	Schwarz criterion	-1.811211	
Log likelihood	51.11726	Hannan-Quinn criter.	-1.994668	
F-statistic	7.762781	Durbin-Watson stat	1.822192	
Prob(F-statistic)	0.000024			

wald test

Wald Test:
Equation: LRFORM_ROW

Test Statistic	Value	df	Probability
t-statistic	1.862180	35	0.0710
F-statistic	3.467714	(1, 35)	0.0710
Chi-square	3.467714	1	0.0626

Null Hypothesis: C(4) = C(5)
Null Hypothesis Summary:

Normalized Restriction (= 0)	Value	Std. Err.
C(4) - C(5)	1.895280	1.017775

Restrictions are linear in coefficients.

Breusch-Godfrey test

Breusch-Godfrey Serial Correlation LM Test:

Null hypothesis: No serial correlation at up to 2 lags

F-statistic	0.316538	Prob. F(2,33)	0.7309
Obs*R-squared	0.790567	Prob. Chi-Square(2)	0.6735

Test Equation:

Dependent Variable: RESID

Method: ARDL

Date: 06/07/26 Time: 09:48

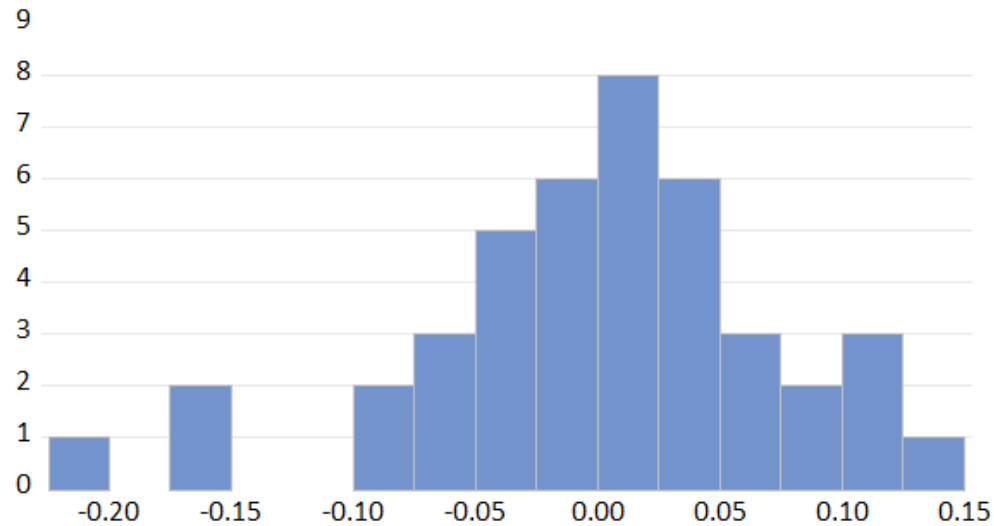
Sample: 1983 2024

Included observations: 42

Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_ROW_TBR(-1)	-0.080454	0.210986	-0.381323	0.7054
LN_ROW_TBR(-2)	0.008198	0.211839	0.038699	0.9694
LN_ROW_TBR(-3)	-0.004871	0.152308	-0.031979	0.9747
LN_NEPAL_GDP	-0.072384	0.304726	-0.237537	0.8137
LN_REER_POS	0.037330	0.802433	0.046521	0.9632
LN_REER_NEG	-0.181598	0.636743	-0.285198	0.7773
C	0.618351	2.712966	0.227924	0.8211
RESID(-1)	0.147965	0.259026	0.571236	0.5717
RESID(-2)	0.085037	0.235195	0.361558	0.7200
R-squared	0.018823	Mean dependent var	2.95E-16	
Adjusted R-squared	-0.219038	S.D. dependent var	0.072515	
S.E. of regression	0.080063	Akaike info criterion	-2.024586	
Sum squared resid	0.211535	Schwarz criterion	-1.652229	
Log likelihood	51.51632	Hannan-Quinn criter.	-1.888103	
F-statistic	0.079135	Durbin-Watson stat	1.979332	
Prob(F-statistic)	0.999569			

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Ramsey RESET

Ramsey RESET Test

Equation: NARDL_ROW

Omitted Variables: Squares of fitted values

Specification: LN_ROW_TBR LN_ROW_TBR(-1) LN_ROW_TBR(-2)

LN_ROW_TBR(-3) LN_NEPAL_GDP LN_REER_POS LN_REER_NEG C

	Value	df	Probability
t-statistic	0.466326	34	0.6440
F-statistic	0.217460	(1, 34)	0.6440
Likelihood ratio	0.267772	1	0.6048

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	0.001370	1	0.001370
Restricted SSR	0.215593	35	0.006160
Unrestricted SSR	0.214223	34	0.006301

LR test summary:

	Value
Restricted LogL	51.11726
Unrestricted LogL	51.25115

Unrestricted Test Equation:

Dependent Variable: LN_ROW_TBR

Method: Least Squares

Date: 06/07/26 Time: 09:49

Sample: 1983 2024

Included observations: 42

Variable	Coefficient	Std. Error	t-Statistic	Prob.
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LN_ROW_TBR(-1)	0.569798	0.302632	1.882810	0.0683
LN_ROW_TBR(-2)	-0.286765	0.189396	-1.514100	0.1392
LN_ROW_TBR(-3)	-0.160094	0.129388	-1.237319	0.2244
LN_NEPAL_GDP	-0.920910	0.401962	-2.291037	0.0283
LN_REER_POS	0.097154	0.837012	0.116072	0.9083
LN_REER_NEG	-2.001424	0.916043	-2.184857	0.0359
C	7.994745	3.616910	2.210380	0.0339
FITTED^2	0.214116	0.459155	0.466326	0.6440
R-squared	0.898905	Mean dependent var	-0.629638	
Adjusted R-squared	0.878092	S.D. dependent var	0.227340	
		Akaike info		
S.E. of regression	0.079377	criterion	-2.059579	
Sum squared resid	0.214223	Schwarz criterion	-1.728594	
Log likelihood	51.25115	Hannan-Quinn criter.	-1.938260	
F-statistic	43.18826	Durbin-Watson stat	1.860141	
Prob(F-statistic)	0.000000			

Heteroscedasticity

Heteroskedasticity Test: Breusch-Pagan-Godfrey

Null hypothesis: Homoskedasticity

F-statistic	0.894196	Prob. F(6,35)	0.5099
Obs*R-squared	5.582469	Prob. Chi-Square(6)	0.4715
Scaled explained SS	5.472058	Prob. Chi-Square(6)	0.4848

Test Equation:

Dependent Variable: RESID^2

Method: Least Squares

Date: 06/07/26 Time: 09:49

Sample: 1983 2024

Included observations: 42

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.142561	0.276632	0.515345	0.6096
LN_ROW_TBR(-1)	-0.025621	0.016018	-1.599527	0.1187
LN_ROW_TBR(-2)	0.015190	0.016309	0.931351	0.3581
LN_ROW_TBR(-3)	-0.026683	0.013616	-1.959684	0.0580
LN_NEPAL_GDP	-0.017647	0.030915	-0.570841	0.5718
LN_REER_POS	-0.063476	0.087448	-0.725869	0.4727
LN_REER_NEG	-0.105848	0.063419	-1.669029	0.1040
R-squared	0.132916	Mean dependent var	0.005133	
Adjusted R-squared	-0.015727	S.D. dependent var	0.008729	
S.E. of regression	0.008798	Akaike info criterion	-6.477660	
Sum squared resid	0.002709	Schwarz criterion	-6.188048	
Log likelihood	143.0309	Hannan-Quinn criter.	-6.371506	
F-statistic	0.894196	Durbin-Watson stat	2.564413	
Prob(F-statistic)	0.509880			

